

Cookbook 1d-discrete	Cookbook 1d-continuous
Cookbook 2d-continous-continuous	Cookbook 2d-discrete-continuous
Cookbook 2d-discrete-discrete	dup_axis() OR sec_axis()
Metamers	Static Graphics (Images)
Aesthetic	geom_point()

• 1d Scatterplot, jitter: <code>geom_point()</code> • 1d Histogram: <code>geom_histogram()</code> • Frequency polygons: <code>geom_freqpoly()</code> • KDE: <code>geom_density()</code> + <code>geom_rug()</code>	• 1d Scatterplot, jitter: <code>geom_point()</code> • Bar charts: <code>geom_bar()</code> • Stacked bar charts: <code>geom_col()</code> • Pie chart/Donut chart/Circular bar chart
• 2d Scatterplot, jitter: <code>geom_point()</code> • Boxplot: <code>geom_boxplot()</code> • Violin plot: <code>geom_violin()</code> • Beeswarm plot: <code>ggbeeswarm::geom_quasirandom()</code>	• 2d Scatterplot, jitter: <code>geom_point()</code> • 2d Histogram: <code>geom_bin2d()</code> • 2d KDE: <code>geom_density2d()</code> • 2d HDR: <code>ggdensity::geom_hdr()</code>
• Creates secondary axes (on top or right of graphic) • Useful in time series	• 2d Scatterplot, jitter: <code>geom_point()</code> • Stacked bar chart: <code>geom_bar()</code> • Mosaic plot • Tree map: <code>treemapify::geom_treemap</code>
• Raster (.png, .jpg): do not scale with data; pixelation on zoom • Vectorized (.pdf, .svg): scales with data; no pixelation on zoom	Different datasets with the same sample statistics
• Makes a scatterplot of points • Point = 0 dimensional object • 0d geom	Visual property of an object in a graphic

<code>geom_path()</code>	<code>geom_line()</code>
<code>geom_segment()</code>	<code>geom_curve()</code>
<code>geom_rect()</code>	<code>geom_tile()</code> OR <code>geom_raster()</code>
<code>geom_polygon()</code>	<code>expression()</code> OR <code>latex2exp::Tex()</code>
<code>base::cut()</code>	<code>facet_wrap()</code>

• Draws lines between successive rows of points AFTER SORTING BY X • 1d geom	• Draws lines between successive rows of points • 1d geom
• Draws curves (curved line segments) • 1d geom	• Draws line segments • 1d geom
• Colored squares on a grid • 2d geom	• Draws rectangles by row • 2d geom
Allows for LaTeX like elements in ggplot labels	• Draws polygons by vertices • 2d geom
Use when faceting by one variable with many sublevels	bins continuous variables into discrete ones

<code>patchwork</code>	<code>ggsave()</code>
<code>theme_set()</code>	<code>theme()</code>
<code>element_text()</code>	<code>theme_update()</code>
<code>guides()</code>	Aesthetic Mapping
Scales	Pairs Plot

Saves ALL plots, not just the last one	Combines plots into one device/window
modify theme elements	• Sets a complete theme • Avoids repeatedly adding + <code>theme_minimal()</code> to your code
Acts like <code>theme.set()</code> but for individual elements	function that modifies text elements of a theme
• Relation between variables and visual elements of the graph • aes: variables $\rightarrow$ aesthetics • (Technically aes: aesthetics $\rightarrow$ variables)	Useful for implementation of multiple legends
• Scatterplot matrix • Use <code>GGally::ggpairs()</code>	• Functions from the data space to the aesthetic space • <code>scale_{aes}</code>: variables sample space $\rightarrow$ range of aesthetic

Parallel Coordinates Chart

Every Plot Demands

Binned Scales

Faceting

Overplotting

Theme

Guides

1. Aesthetic Mapping 2. Layers composed of geoms and stat transformations 3. Scales 4. Coordinate System 5. Facet Specification 6. Theme	• Used to visualize higher dimensional data • Typically used to identify clusters in data • Use <code>GGally::ggparcoord()</code>
• Breaks a plot into several plots each using a different subset of data • Use <code>facet_grid()</code> or <code>facet_wrap()</code>	Converts continuous variables into discrete ones
Collection of non-aesthetic specifications that qualify how a plot is displayed	• Visual confusion in a graphic caused by plotting too much data • Remedies: 1. Jittering 2. Shrink points 3. Alpha blending
	Determine how scales are presented